

Review Article

Electrochemical detection of nitrate, nitrite and ammonium for on-site water quality monitoring

Jing Fang Tan¹, Amie Anastasi^{1,2} and Shaneel Chandra^{1,2}**Abstract**

This review examines the most recent electrochemical developments for nitrate, nitrite and ammonium detection for on-site water monitoring. There remains a high demand for effective field-based detection of the dissolved inorganic nitrogen (DIN) analytes to aid in mitigating nitrogen loading. Electrochemical approaches show increasing potential to fill this role as advancements in nanotechnology continually improve analytical performance and on-site applicability. However, translating these improvements into the field still faces the resonating challenges of reaching analytical proficiency (selectivity, sensitivity, robustness, stability), practical end-user functionality, minimal matrix interferences and cost effectiveness. Herein, we elaborate on these challenges via a critical evaluation of current studies and examine how realistic the prospects of on-site nitrate, nitrite and ammonium are. We also present recommendations in addressing these gaps to conclude the review.

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Introduction

The over-accumulation of nitrogen species, primarily the dissolved inorganic nitrogen (DIN) ions (nitrate, nitrite and ammonium) in aquatic systems by anthropogenic activities is widely recognised as a significant

environmental pollutant [1]. Within Australia, all sites that release to receiving waters must implement programs to monitor and assess DIN discharges [2]. These programs seek to identify and ensure that emerging issues can be addressed if they arise, so that conservation of Australia's coastal ecosystems, such as the Great Barrier Reef, will persevere for future generations.

Currently, the two dominant methods for DIN monitoring involve either the collection and transportation of surface water samples to an offsite laboratory, or the use of automated real-time spectrophotometric sensors based on site [3,4]. The traditional lab-based methods are time and labour intensive, often requires large sample volumes and can risk sample integrity. In contrast, the on-site spectrophotometric devices provide rapid and frequent data that reduces the time lag of implementing countermeasures at point and time of high loading events [3]. However, one drawback for these field-based UV-spectrophotometric devices are their high cost. They are commonly based on complex detection platforms that require skilled personnel, lacks longevity, and necessitates probe replacements every few months [3,4].

New discoveries and developments continually introduce improvements to monitoring processes. There has recently been a noticeable global shift in attention toward electrochemical sensor devices. Rapid advancements in novel nanomaterials in the last decade have shown substantial enhancements in electrochemical performance of target analytes, including the DIN species. Miniaturisation of nano- and micro-sized transducer components provide flow-on benefits such as portability, rapid real-time measurements, lowered operating costs of equipment and reagents and reduced chemical footprint. In addition, handheld potentiostats have progressed to on-site operations with the ability for data acquisition and relay, improving practicability in remote locations [5].

While the progress in electrochemical technologies demonstrate significant potential for commercialisation, DIN electrochemical sensors have yet to show widespread success as a marketable device. There are many reviews documenting the progress in DIN sensor research, however, few report comparisons of application beyond ideal laboratory conditions [6,7]. This represents a large gap between proof of concept and its

successful demonstration in the field. To address this gap, we present this mini review exploring the feasibility of sensors reported in the literature for on-site monitoring of nitrate, nitrite and ammonium individually based on current research. It is hoped that this work will provide a guiding snapshot of current challenges to electrochemical DIN monitoring for others.

Nitrate analysis

Nitrate is the most abundant of the DIN species in unpolluted water systems. Under oxygenated conditions, ammonium and nitrite undergo biological nitrification to produce nitrate [8]. The nitrate molecule is highly labile and leaches readily through the soil from agricultural sites, particularly in wet conditions [9]. Recent research progress for electrochemical nitrate analysis has predominantly been voltammetric/ampereometric and potentiometric measurements under ideal laboratory conditions.

Voltammetry/ampereometry

Comparative studies done on nitrate reduction using different polycrystalline electrodes show that reaction kinetics generally improved in the presence of the transition metals toward the top left of the *d*-block in the periodic table ($\text{Cu} > \text{Ag} > \text{Au}$ for the coinage metals and $\text{Rh} > \text{Ru} > \text{Ir} > \text{Pd/Pt}$ for the transition metals), arising from the greater capacity of these metals to adsorb species such as nitrate [10]. Various metal cathode surfaces as well as those based on carbon or its various forms such as graphene, mesoporous carbon and activated carbon have been extensively investigated for electrochemical analysis of nitrate [11,12]. Among all of these, Cu was the most studied due to its higher efficiency and low cost [13].

Voltammetric detection of nitrate using Cu have involved innovative material composites and immobilisation techniques with varying degrees of success. Cu nanowire growth in the pores of a polycarbonate membrane was shown to significantly increase electrode surface area and consequently enhance the kinetics of the modified electrode, reaching detection limits of $9.1 \mu\text{M}$, significantly lower than bare Cu electrodes [11]. Other designs include ion imprinted Cu nanoparticles-polyaniline nanocomposite electrode [14] and Cu with TiO_2 core shells bound with Nafion® and polyalizarin modified electrodes [15], with detection limits of $5 \mu\text{M}$ and $2.1 \mu\text{M}$ respectively.

Challenges associated with the use of Cu electrodes include the requirement of highly acidic or alkaline electrolyte to facilitate the electrocatalytic reduction of nitrate [16]. In acidic solutions, care must be taken to avoid parasitic hydrogen evolution from an overabundance of H^+ ions [11]. Such precise pH maintenance complicates applicability of Cu based electrodes

in field settings. Another limitation is the prevalence of Cl^- ions in all natural waters. Cl^- ions, at concentrations as low as 10 ppm, cause suppression of nitrate measurements, likely due to the formation of CuCl on the sensor surface [11,17]. While the Cl^- effect can be mitigated by adding silver sulphate (Ag_2SO_4) to the sample in order to precipitate out as AgCl , it is impractical in remote surroundings [11].

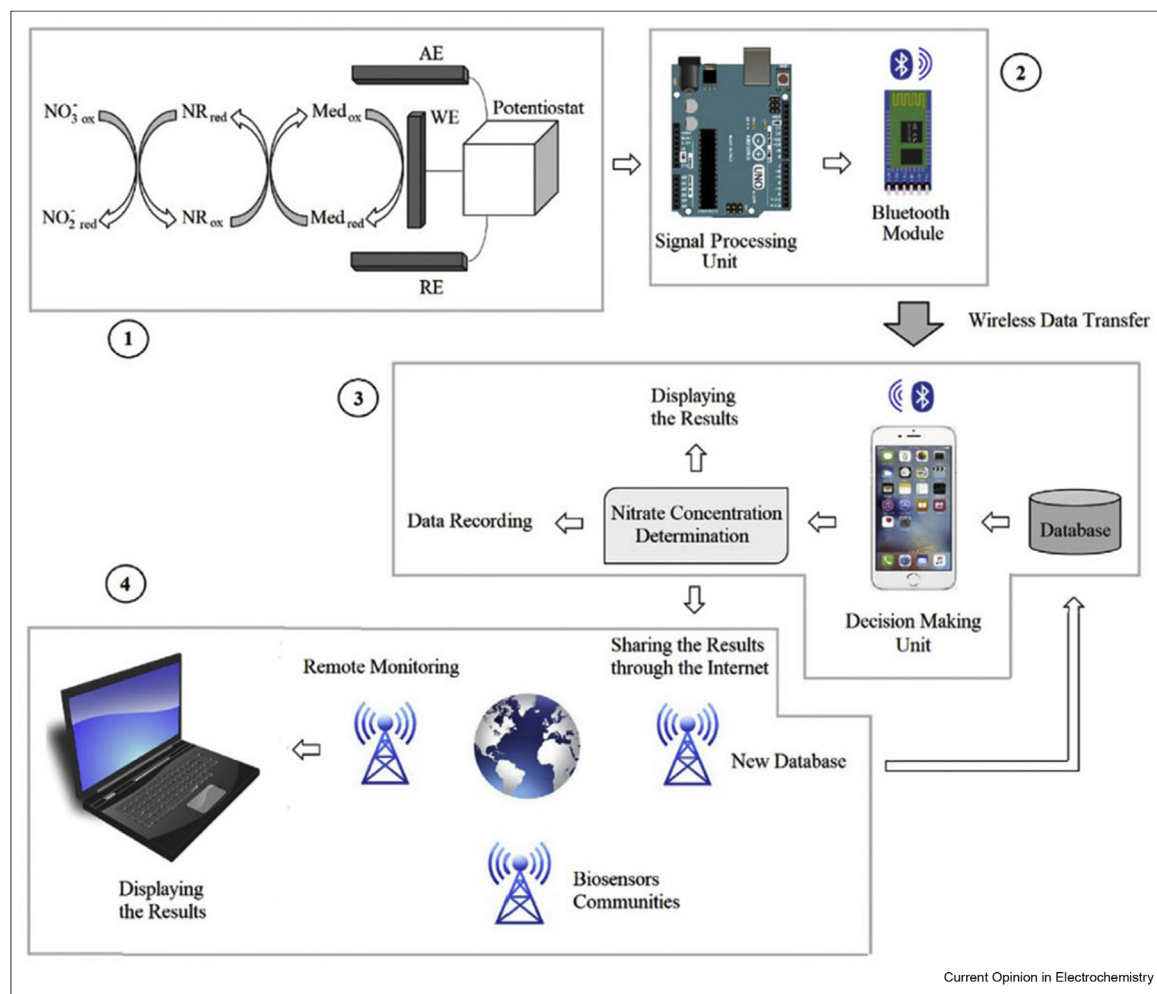
Biosensors for nitrate analysis involve scaffolding of nitrate reductase (NaR) onto electrode surfaces, imparting a far higher degree of selectivity than metal cathode sensors [18]. Challenges associated with NaR electrodes include complex preparation techniques leading to inconsistent bonding strength and electrode lifespans [18,19*]. Enzymes are known to easily denature in response to pH and temperature, which limits electrode longevity. The presence of dissolved oxygen in samples also causes interference at low reduction potentials and reacts readily with mediators required for enzymatic function [20]. In lab-based methods, oxygen is commonly removed by purging the samples with nitrogen gas, a non-viable process in field settings.

To address oxygen interferences, chemical oxygen scavengers have been tested as an alternative to gas-purging. Sodium sulfite, an antioxidant, was trialled in conjunction with the biosensing of nitrate, with minimal efficacy [20]. Glucose oxidase was also assessed, and it was found that with the catalyst pyranose 2-oxidase, oxygen was successfully removed from samples for more accurate nitrate detection [21]. A drawback to using oxygen scavengers are the complications of adding further reagents for on-site testing [20].

To factor in NaR enzyme's natural decline in activity on the sensor surface, Massah et al. introduced an intelligent biosensor to detect nitrate with a support vector machine learning in order to predict the activity decrement over time [19*]. In this manner, the lifespan of the sensor was prolonged to beyond 10 days without requiring frequent enzyme replacement, whereas other NaR electrodes reported an almost immediate decline in useability [22]. The device used disposable screen-printed electrodes (SPEs) and a portable potentiostat incorporated with wireless data transfer unit and internet of things (IoT) based cloud server unit (Figure 1). This design is an example of how artificial intelligence, though not directly involved in the electroanalytics, can help to compensate for changing environmental conditions accompanying field analysis.

An advantage of voltammetry over other electrochemical techniques is the ability to detect multiple analytes on the same electrode when their oxidation or reduction potentials are distinct from one another, resolvable on the potentiostat and when interferences are minimal. Many studies have reported sensors (both enzymatic

Figure 1



Schematic for nitrate biosensor, 1) electrochemical biosensing (AE – Auxiliary electrode, WE – Working electrode, RE – Reference electrode) 2) signal processing and wireless transfer 3) decision making unit 4) IoT cloud server. Image modified from Massah *et al.* [19^{*}] with permission from Elsevier.

and non-enzymatic) that are capable of detecting two of the DIN analytes [12,23,24].

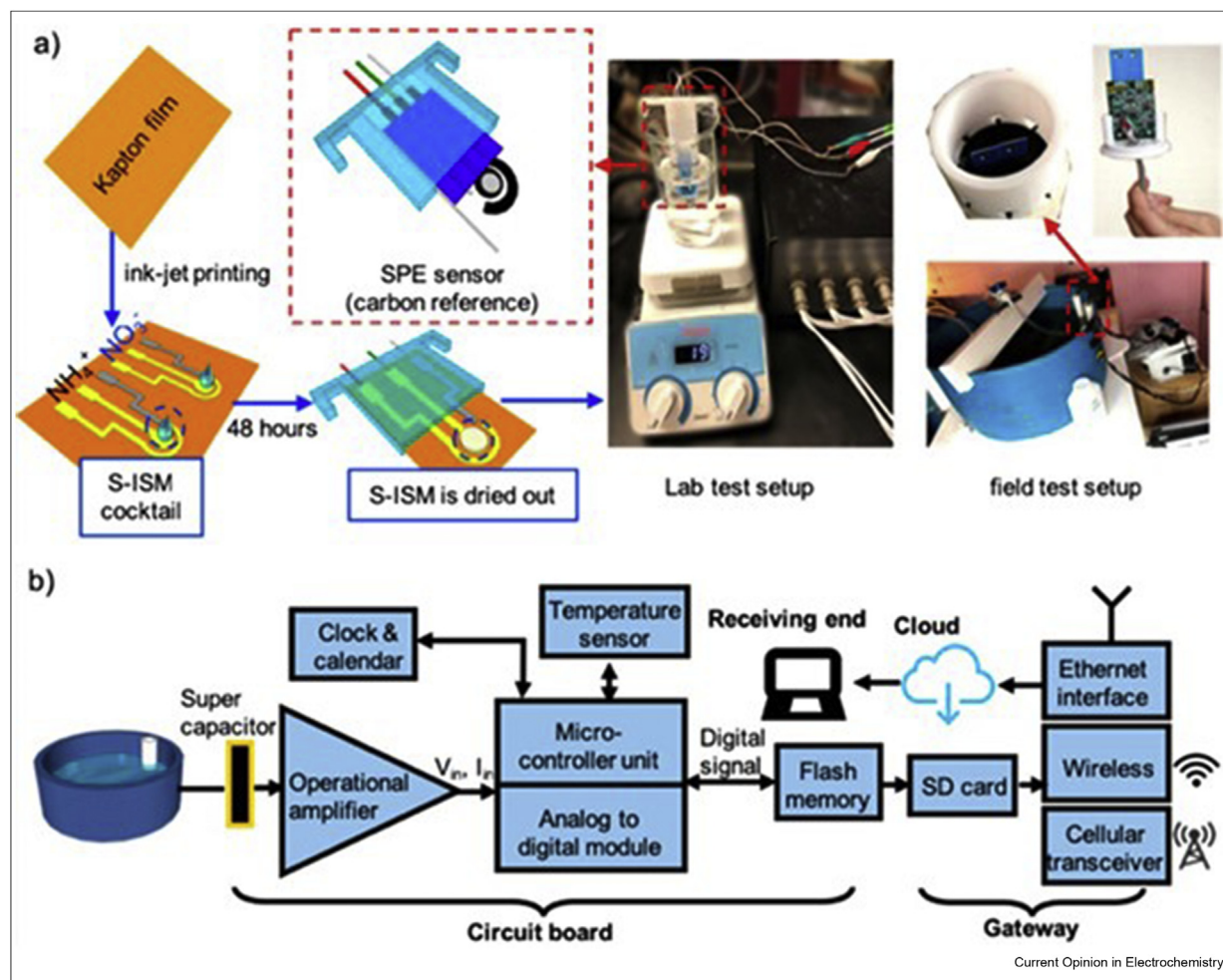
Potentiometry

Ion selective electrodes (ISEs) have greater simplicity in terms of their electric circuitry and consequently allow faster reactions at lower cost. Due to this and their ease of use, ISEs have shown the most success in the market. Recognised weaknesses for ISEs, however, include precision, accuracy, robustness and sensitivity [25]. It is typical for potentiometric analysis to suffer from drift; this is a trait within all the ISE's reviewed here and in commercial nitrate ISE's [26–28]. End-users are made aware of their associated limitations before usage. The issue of drift is addressed by conducting calibrations

before each usage, requiring additional chemicals and time when enacting field measurements [27].

Algorithmic corrections are commonly used to control some of the known inconsistent variables for ISE systems [29]. For instance, an SPE modified by drop-coating the working electrode with a nitrate ionophore (9-hexadecyl-1,7,11,17-tetraoxa-2,6,12,16-tetraazacycloicosane) to create a solid-state ion selective membrane (ISM) electrode was imbedded with an autocorrection unit to compensate for temperature influences [27]. A temperature sensor was attached to the portable nitrate ISE and the autocorrection unit, and wireless data transfer allowed real-time measurements into a cloud server (Figure 2). The temperature compensation equation

Figure 2



(a) Fabrication of ion-selective membrane sensors and test setup. (b) Data transmission process of the wireless ion selective-membrane sensors. Image modified from Huang et al. [27] with permissions from ACS Publications.

corrected up to 19.6% error compared to when no temperature compensation was used [27].

An electrode modified with the same ionophore (9-hexadecyl-1,7,11,17-tetraoxa-2,6,12,16-tetraazacycloicosane) for nitrate analysis additionally immobilised polytetrafluoroethylene (PTFE) within the ion selective membrane in an attempt to increase the antifouling ability [30**]. By increasing the hydrophobicity of the surface, this extra layer alleviated biofouling and thus enhanced the robustness of the electrode.

Biofouling is a major detrimental factor to consider for commercialisation of any field sensor. Few literature studies for electrochemical nitrate detection have focused on extending in-situ application periods by imparting antifouling ability.

Table 1 lists different electrochemical sensors for nitrate analysis along with their analytical features (analytical method, limit of detection, sensitivity, dynamic linear range and sample matrix).

Nitrite analysis

The monitoring of nitrite is of particular interest in drinking water and food due to its higher relative toxicity to humans. High nitrite concentrations in humans are known to be associated with methemoglobinemia (blue baby syndrome) and stomach cancer [32].

The partially oxidized form of nitrogen in nitrite allows the dual versatility of its detection based on either oxidation or reduction [33]. The electrochemical oxidation of nitrite offers several advantages as only one product (nitrate) forms and the detection will not suffer

Table 1

Comparison among electrochemical sensors for nitrate detection.

Electrode	Detection Method	Limit of Detection (μM)	Sensitivity (as reported)	Linear Range (as reported)	Sample Matrix	Ref.
Cu nanowires on polycarbonate membrane	LSV	9.10	$0.0636 \mu\text{A } \mu\text{M}^{-1} \text{ cm}^{-2}$	10–50 μM	water	[11]
Cu_xO ($x=1$ or 2) impregnated glassy carbon spheres–basal plane pyrolytic graphite electrode	CV	0.0170	$1650 \mu\text{A } \mu\text{M}^{-1} \text{ cm}^{-2}$	10–100 μM	soil, fruits	[12]
Ion imprinted polymer using Cu nanoparticles – polyaniline nanocomposite	LSV, CV	5.00	$0.00088 \mu\text{A } \mu\text{M}^{-1} \text{ cm}^{-2}$	1–100 μM	water	[14]
Cu on TiO_2 coreshell – nafion - polyalizarin modified electrode	DPV, CV	2.1	$0.0123 \mu\text{A } \mu\text{M}^{-1}$	5–7500 μM	water	[15]
Intelligent nitrate reductase on glassy carbon electrode with wifi	LSV	unspecified	unspecified	0–2000 μM	water, juice	[19*]
Ag/Pt dual region on indium tin oxide glass electrode	DPV	134	unspecified	unspecified	water	[23]
Wireless ion selective electrode with temperature compensation	Potentiometry	16.1	48.9 mV/decade $[\text{NO}_3^-]$	1–64 ppm	water	[27]
Poly(tetrafluoroethylene) (PTFE) - solid state ion selective screen-printed electrode	Potentiometry	16.1	58 mV/decade $[\text{NO}_3^-]$	1–64 ppm	water	[30**]
Au nanoparticles on Pd modified electrode	LSV	1.19	$4.7 \mu\text{A mg}^{-1} \text{ L}$	0.2–15 ppm	water	[31]

Acronyms: CV – cyclic voltammetry, LSV – linear sweep voltammetry, DPV – differential pulse voltammetry.

Note: Limit of detection (LOD) converted to μM for consistency.

from interference of dissolved oxygen. Both oxidation and reduction pathways have been explored using enzymatic and non-enzymatic approaches, leading to the immense diversity of nitrite electrochemical assay types [22,34–36]. Only voltammetric/amperometric techniques for nitrite analysis will be examined as it is the most prominent in recent studies.

Voltammetry/amperometry

The interest in nitrite determination in the medical field as well as its electrochemical diversity, has adapted a rising research trend of greater innovative and greener methods for electrode fabrication [34–38]. Majari *et al.* developed an eco-friendly synthesis using *Acacia caesia* flower extract to synthesise Ag–Cu nanoparticles and enhance the binding capability of the Ag–Cu to ZnO [35]. Another example discovered that carbon fibre paper treated with simple air annealing led to high-performance nitrite detection [37].

An SPE onto which Au nanoparticle, NiO nanoparticles and rGO were electrochemically deposited was interfaced with a smartphone-based electrochemical system, and along with GPS coordinates, was used to detect nitrite levels in different locations [39**]. On-site procedures involved filtering and mixing the samples with 0.1 M phosphate buffer (supporting electrolyte) before administering the sample to the sensor. The study shows quantitation results and geospatial information uploaded through a mobile network of smartphones at different locations (Figure 3) [39**].

Nitrite biosensor fabrication studies have commonly utilised nitrite reductase (NiR) enzymes. Similar to nitrate reductase (NaR), limitations such as the interference from dissolved oxygen is a hindrance to successful measurement. In an attempt to avoid the dissolved oxygen interference, a study immobilised glucose oxidase and catalase with cytochrome *c* NiR during the electrode surface deposition in order to create anoxic conditions when detecting nitrite in urine [40]. This mechanism of using glucose oxidase to remove oxygen is similar to the trials for nitrate detection done by Plumere *et al.*, the main difference is that enzymes were immobilised directly onto the electrode surface, improving the electrochemical process [21].

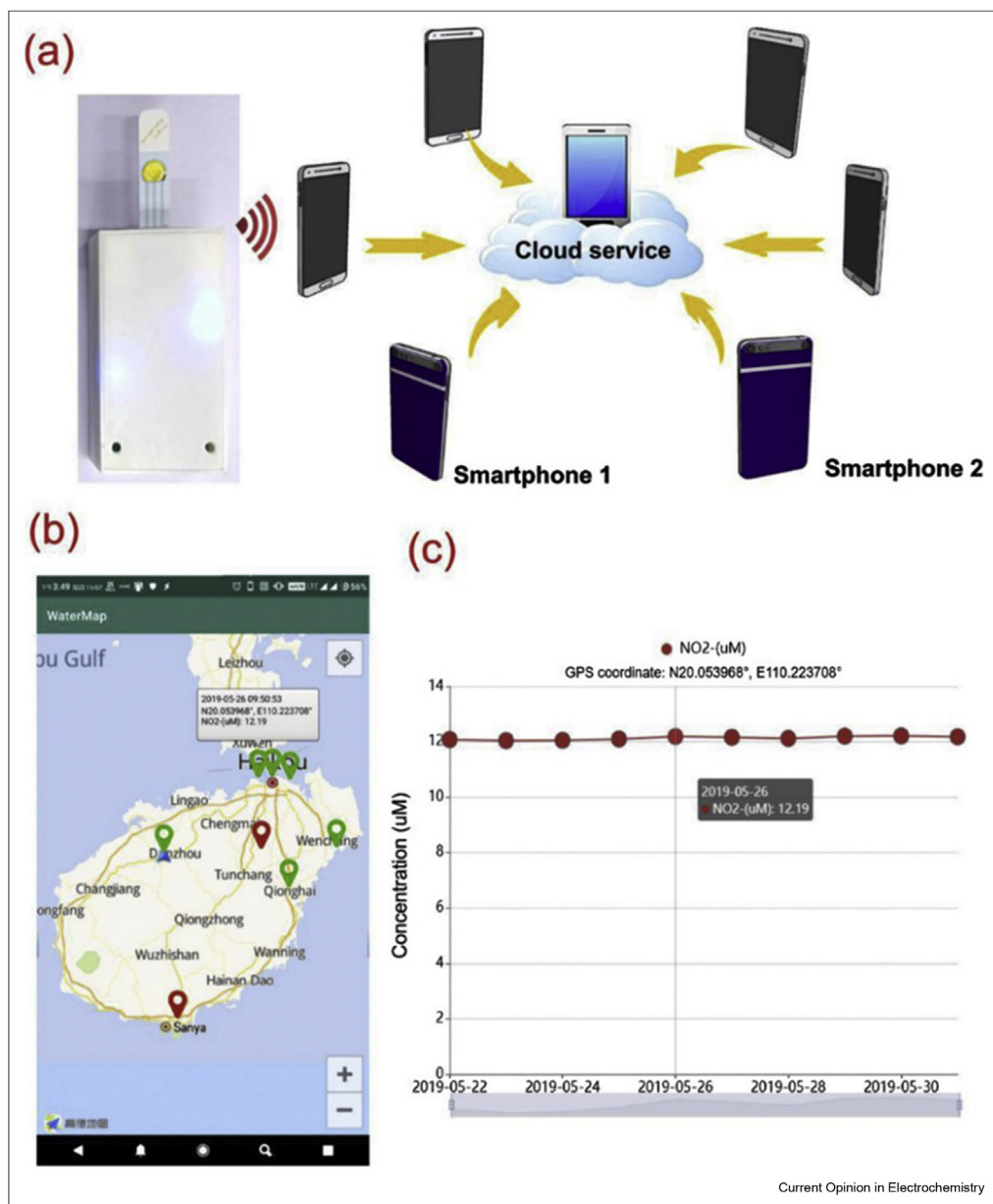
The rich research of using biological reagents shows potential for eliminating the influence and negative consequences of chemical reagents. However, none have reached maturity for practicality, and many require additional steps in the extraction of substances [41].

Table 2 lists different electrochemical sensors for nitrite analysis along with their analytical features (analytical method, limit of detection, sensitivity, dynamic linear range and sample matrix).

Ammonium analysis

Ammonium is the most reduced form of nitrogen in aquatic systems. At higher pH, the formation of ammonia predominates, which poses a higher toxicity to aquatic organisms [43].

Figure 3



(a) Gathering information of nitrite levels from different locations to a cloud server. (b) Spatiotemporal mapping using GPS. (c) Tracking the nitrite levels over time. Image adopted from Xu et al. (2020) with permission from Elsevier [39**].

Voltammetry/amperometry

Despite the difficulty of direct voltammetric measurements due to the slow electron transfer rate of ammonium on electrodes, there have been several studies using this technique [23,44,45]. A study successfully out-manoeuvred this limitation by creating a nano-structured morphology of polyaniline:polystyrene – sulphonate (nano-PANI:PSS) that allowed selective insertion of ammonium into the nano-PANI:PSS matrix [45]. The insertion process (‘doping’) occurs during

reduction, subsequent oxidation releases the ammonium (‘de-doping’) from the polymer matrix and results in well-defined peaks proportional to ammonium concentration [45].

Ammonium biosensors use enzymes as transducers to measure ammonium indirectly [41]. Typical enzymes used for ammonium biosensors are glutamate dehydrogenase (GLDH) or alanine dehydrogenase (AlaDH) [41]. Ammonium is indirectly measured from the

Table 2

Comparison among electrochemical sensors for nitrite detection.

Electrode	Detection Method	Limit of Detection (μM)	Sensitivity (as reported)	Linear Range (as reported)	Sample Matrix	Ref.
Cu_xO ($x=1$ or 2) impregnated glassy carbon spheres–basal plane pyrolytic graphite electrode	CV	0.0330	$1500 \mu\text{A } \mu\text{M}^{-1} \text{ cm}^{-2}$	$0.1\text{--}10 \mu\text{M}$	soil, fruits	[12]
CuO - protonated carbon nitride - reduced graphene oxide on glassy carbon electrode	CV, EIS	0.0160	$0.093 \mu\text{A } \mu\text{M}^{-1}$	$0.2\text{--}110 \mu\text{M}$	water	[32]
Ag–Cu decorated ZnO nanocomposite on glassy carbon electrode	CV	17.0	unspecified	$100\text{--}1500 \mu\text{M}$	water	[35]
Oxidised carbon fibre paper	Amperometry	0.0700	$950.4 \mu\text{A } \text{mM}^{-1} \text{ cm}^{-2}$	$0.1\text{--}3838.5 \mu\text{M}$	water, sausages	[37]
Au nanoparticles, NiO nanoparticles and reduced graphene oxide (rGO) on screen printed carbon electrodes (SPCE)	CV, DPV	0.2	unspecified	$1\text{--}500 \mu\text{M}$	water	[39**]
Cytochrome c nitrite reductase screen printed electrode with glucose oxidase and catalase	CV	4.70	$326 \text{ mA } \text{M}^{-1} \text{ cm}^{-2}$	$2\text{--}200 \mu\text{M}$	urine	[40]
Cu ions-crosslinked bovine serum albumin nanoflower networks on glassy carbon electrode	DPV	0.100	unspecified	$0.5\text{--}500 \mu\text{M}$	water	[42]

Acronyms: CV – cyclic voltammetry, DPV – differential pulse voltammetry, EIS – electrochemical impedance spectroscopy.

Note: Limit of detection (LOD) converted to μM units for consistency.

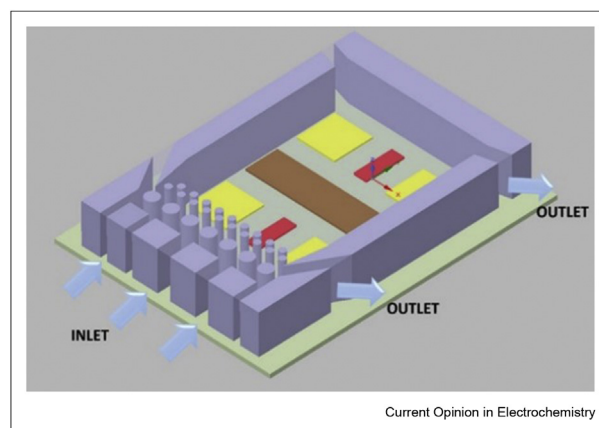
oxidation of NADH to NAD^+ [46]. Biological enzyme detection of ammonium exhibits good selectivity and high sensitivity with the ability to be adapted; it is a promising pathway but has yet to reach a maturity for practicable application [41]. The general challenges encountered with enzyme usage, such as denaturation, complex preparation, inconsistent immobilisation, and short lifespan still require refinement.

Potentiometry

The bulk of electrochemical research for ammonium detection is done using potentiometric ISE's with nonactin as the conventional ionophore. pH electrodes are also frequently attached to ammonium sensors to elucidate the ammonium/ammonia ratio [6,47]. However, in the presence of K^+ and Na^+ ions, nonactin shows poor selectivity for ammonium due to the similarity in ionic size and monovalent charge between all three cations [6]. This is why ammonium detection in seawater or estuarine environments using nonactin-based sensors tend to be problematic [6]. There have been several studies attempting use of alternative ionophores, however the interfering ions, particularly K^+ remain an issue [6].

A novel sensor for field analysis used a microfluidic lab-on-a-chip for real-time ammonium detection in sewage applications [48**]. The microfluidic system was engineered so that the detection microchamber had minimal exposure to the environment, thus allowing a degree of protection against marine biofouling (Figure 4). Potentiometric readings were comparable to laboratory studies

Figure 4



Assembly of a PDMS-microfluidic lab-on-a-chip with silicon transducer. Image modified from Gallardo-Gonzalez *et al.* [48**] with permissions from Elsevier.

using nonactin and testing under real sewage conditions resulted in stable readings at the electrodes for at least 15 min [48**].

There are commercial ammonium ISE's available for purchase, of which possess limitations typical of potentiometric measurements [49].

Table 3 lists different electrochemical sensors for ammonium analysis along with their analytical features

Table 3

Comparison among electrochemical sensors for ammonium detection.

Electrode	Detection Method	Limit of Detection (μM)	Sensitivity (as reported)	Linear Range (as reported)	Sample Matrix	Ref.
Ag/Pt dual region on indium tin oxide glass electrode	DPV	3.95	unspecified	unspecified	water	[23]
Ni foam supported Pt nanoparticles - Ag/polypyrrole electrode	LSV, CV	0.037	0.089 mA	0.1–100	water	[44]
Polyaniline:polystyrene – sulphonate electrode	SWV, CV	26.9	106 mA M cm	0.1–11500	water	[45]
Microfluidic lab on a chip	Potentiometry	40.0	unspecified	unspecified	water	[48**]
Poly(3,4-ethylenedioxythiophene): Poly(styrenesulfonate) (PEDOT:PSS) transducer with Poly(o-phenylenediamine) (PoPD)-PEDOT:PSS ion selective membrane screen printed electrode	Potentiometry	57.0	61.9 mV/decade $[\text{NH}_4^+]$	1–1000 mM	water	[50]

Acronyms: CV – cyclic voltammetry, LSV – linear sweep voltammetry, DPV – differential pulse voltammetry, SWV – square wave voltammetry.

Note: Limit of detection (LOD) used μM for consistency.

(analytical method, limit of detection, sensitivity, dynamic linear range and sample matrix).

Conclusion and prospects

In recent years, new gains such as enhanced nano-material surface kinetics, improved computer learning, higher anti-fouling ability and convenient portability have been achieved for the electrochemical detection of DIN analytes. Continued research will no doubt further invigorate the possibilities for on-site capabilities, the future of electro-analytics for DIN detection is bright. Main factors to consider in extending these exciting innovations to realistic operations in the field are the challenges of obtaining the required analytical proficiency (selectivity, sensitivity, robustness, and stability), practical end-user functionality, minimal matrix interferences and cost-effectiveness. Work remains to bridge the proof of concepts addressing these challenges to achieve effective on-site practicality for each of the DIN analytes.

A mismatched level of emphasis is placed on superior laboratory performances such as ultralow limits of detection and miniaturisation rather than the functionality and ease of implementation carried out in the field. Therefore, an integrated monitoring network between the scientific and management communities would be the most beneficial way to accelerate sensor development. A key example of this is the collaboration of researchers and industries in the hopes of improving DIN monitoring in waterways surrounding the environmentally sensitive areas of the Great Barrier Reef.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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